EXAMPLE 6.15

If the focal length is 150 cm for a lens, what is the power of the lens?

Solution

Given, focal length, $f = 150\ \text{cm} = 1.5\ \text{m}$

Equation for power of lens is, $P = \frac{1}{f}$

Substituting the values,

$$P = \frac{1}{1.5\,\text{m}} = 0.67\,\text{D}$$

As the power is positive, it is a converging lens.

6.6.6 Focal length of lenses in contact

Let us consider two lenses ① and ② of focal lengths f_1 and f_2 placed coaxially in contact with each other so that they have a common principal axis. For a point object placed at O beyond the focus of the lens ① on the principal axis, an image is formed by it at I'. This image I' acts as an object for the lens ② and the final image is formed at I as shown in Figure. 6.37. As these two lenses are thin, the measurements are done with respect to the common optic centre P between the two lenses.

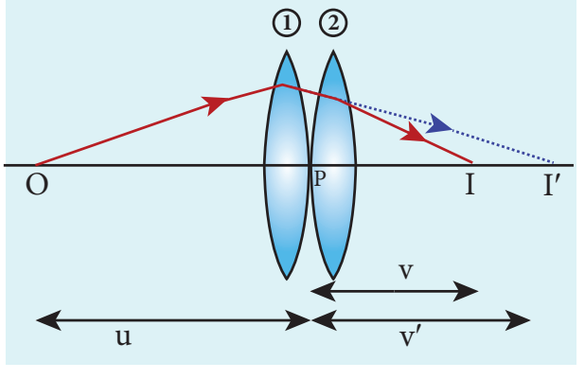


Figure. 6.37 Lenses in contact

For the lens ①, the object distance PO is u and the image distance PI' is v'. For the lens ②, the object distance PI' is v' and the image distance PI is v.

Writing the lens equation (6.63) for lens ①,

$$\frac{1}{v'} - \frac{1}{u} = \frac{1}{f_1} \qquad (6.70)$$

Writing the lens equation (6.63) for lens ②,

$$\frac{1}{v} - \frac{1}{v'} = \frac{1}{f_2} \qquad (6.71)$$

Adding the above two equations (6.70) and (6.71) gives,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \qquad (6.72)$$

The combination acts as a single lens of focal length f so that for an object at the position O, it forms the image at I. Then,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad (6.73)$$

Comparing the equations (6.72) and (6.73) we can write,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \qquad (6.74)$$

The above equation can be extended for any number of lenses in contact as,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \frac{1}{f_4} + \ldots\ldots\ldots \qquad (6.75)$$

The above equation can be written in terms of power of the lenses as,

$$P = P_1 + P_2 + P_3 + P_4 + \ldots\ldots\ldots \qquad (6.76)$$

Where, P is the net power of the lenses in contact. One should note that the sum in equation (6.76) is an algebraic sum. The powers of individual lenses may be positive (for convex lenses) and negative (for concave lenses). Combination of lenses helps to obtain converging (or) diverging lenses of desired magnification. Also, combination of lenses enhances the sharpness of the image. As the image formed by the first lens becomes

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